

REQUIREMENT GATHERING

Date: 20/07/2026

1. Project Overview:

MediSense is an AI-powered healthcare monitoring system designed to tackle the critical issue of medication non-adherence in chronic patients (specifically those with Type-2 Diabetes and Hypertension). The system tracks daily medication compliance and utilizes machine learning models to analyze patterns of missed doses, enabling early prediction of potential health crises and providing proactive dashboards and alerts for healthcare professionals.

2. System Scope:

- The scope of this proposed system is an academic prototype and research project for the MCA curriculum. It aims to demonstrate the feasibility of combining daily patient medication self-logging with background risk prediction models using a clean, scalable tech stack.

3. Target Audience:

- The system involves two primary user groups:
 1. **Patients:** Individuals diagnosed with Type-2 Diabetes or Hypertension who need an easy-to-use interface to track and log their daily medication schedules.
 2. **Doctors/Medical Professionals:** Medical practitioners who require an administrative dashboard to monitor patient compliance data and receive automated high-risk alerts.

4. Modules:

The core architecture of the system consists of the following key modules:

- **Authentication & User Management Module:** Handles secure login, registration, and role assignment for admins, doctors, and patients.
- **Patient Medication Logging Module:** Allows patients to view their prescribed medical regimens and log daily dosage consumption.

- **Doctor's Dashboard Module:** Provides clinical views of patient lists, adherence statistics, and real-time alerts.
- **AI Risk Prediction Module:** Runs predictive models in the backend using historical compliance data to flag patients moving into critical risk windows.

5. User Roles:

- **Administrator (Admin):** Full system access to manage hospital setups, register authorized doctors, and oversee system logs.
- **Doctor:** Access to patient health records, medication assignment tools, and adherence alert dashboards.
- **Patient:** Restricted access limited strictly to viewing personal prescriptions, daily schedules, and self-logging their dose consumption.

6. System Ownership:

- The system ownership belongs to Amal Jyothi College of Engineering, Kanjirapally as an Academic Project submission.

7. Industry/Domain:

- Healthcare and Medical Informatics

8. Data Collection Contacts:

- Name: Dr. Vijinla Vincent
- Role: Assistant Surgeon
- Contact Information: CHC, Ezhumattoor

Ph No: 9744553202

9. Questionnaire for Data Collection:

- **Domain:** Clinical Practice & AI-Assisted Patient Adherence
- **Respondent:** Dr. Vijnla Vincent, Assistant Surgeon

Q1. How common is medication non-adherence among patients diagnosed with Hypertension and Type-2 Diabetes in your daily practice?

Response: In daily clinical practice, non-adherence is extremely prevalent, affecting approximately **70% of diagnosed patients**. A primary reason for this high rate is symptom-driven medication taking—patients frequently pause or stop their prescribed medication once they feel asymptomatic, resuming it only when physical discomfort or symptoms resurface.

Q2. What are the most frequent reasons patients provide for missing or skipping their prescribed doses (e.g., forgetfulness, lack of symptoms, side effects)?

Response: Apart from general forgetfulness and asymptomatic relief, a major factor is the **fear of potential side effects**. Many patients prematurely stop their dosages out of apprehension regarding long-term organ impact or immediate discomfort. However, patients often fail to realize that the clinical risks and side effects of uncontrolled Hypertension and Diabetes are far more severe and damaging than the mild, manageable side effects of prescribed pharmaceuticals.

Q3. When a patient misses their BP or Diabetes medication consecutively for 2 to 3 days, what are the immediate and long-term clinical risks they face?

Response: Missing doses for 2 to 3 days rarely leads to immediate, catastrophic acute events for most stable chronic patients. However, persistent non-adherence over 1 to 2 months or longer poses severe long-term clinical risks, including:

- **Cardiovascular Complications:** Increased risk of myocardial infarction (heart attack), hypertensive heart disease, and stroke.

- **Renal Damage:** Diabetic and hypertensive nephropathy leading to chronic kidney disease (CKD).
- **Microvascular & Neurological Damage:** Diabetic retinopathy (vision impairment/blindness) and peripheral neuropathy.
- **Uncontrolled Spikes:** Severe glycemic variability and rebound hypertension.

Q4. Is there currently any systematic method or software used in your clinic/hospital to track whether a patient is taking their medications regularly after leaving the hospital?

Response: No, there is currently **no dedicated software or systematic tracking mechanism** deployed at the clinical level to monitor post-discharge adherence.

Q5. If my system detects that a patient has consistently missed multiple doses, how helpful would a predictive dashboard alert be for you to intervene before a medical emergency occurs?

Response: Such a system would be **highly valuable and impactful**. Due to demanding clinical schedules, doctors cannot manually follow up with every patient individually. Automated predictive alerts bridge this gap. However, for maximum clinical utility, the system must remain concise and streamlined—if patient volume is extremely high, doctors will have limited time to process complex interfaces. The alerts must be actionable and prioritized.

Q6. What clinical parameters or risk indicators (e.g., age, history of stroke, number of missed doses) should our AI model prioritize to categorize a patient as 'High Risk'?

Response: To effectively stratify patients into a 'High Risk' category, the AI model should prioritize and weight the following key clinical parameters:

1. **Cumulative Missed Doses:** Consecutive missed medication logs exceeding 7+ days within a 30-day window.

2. **Comorbidities & Critical Medical History:** Prior history of stroke, transient ischemic attack (TIA), heart attack, or underlying Chronic Kidney Disease (CKD).
3. **Age & Demographics:** Advanced age (60+ years), where missing doses accelerates vascular complications.
4. **Baseline Biomarkers:** Poor baseline control indicated by elevated HbA1c levels (>8.5%) or persistent stage-2 hypertension.

Q7. Do you think a patient-facing web/mobile application where they can self-log their daily medication intake would be practical and useful?

Response: Of course it would be practical and useful for patients who regularly use web and mobile apps.

Q8. If a patient fails to log their medication, would sending automated reminders via push notifications or offline SMS or IoT based alarms significantly help in improving adherence?

Response: It would be more helpful if IoT based alarms was implemented as it helps to notify patients to update their daily doses details in the portal even though they are offline.

Q9. In the event of a high-risk alert generated by the system, which communication channel would you prefer for notification (such as an in-app dashboard alert, IoT based alarm, or a direct SMS)?

Response: In case of a high-risk alert, I would prefer an IoT based or direct SMS alert for notifying the patient as it gives an instant and offline mode of alarm or notification to the patient reminding to take their dosages

Q10. What major challenges or resistance do you foresee from both doctors and patients when implementing a digital medication tracking and risk prediction system like this?

Response:

- It might be challenging for elderly patients and those who are not keen to use such technologies.
- Need technical training

10. Geotagged Photos



